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THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING

Daily 2026-10-31 Ivanti Connect Secure CVE-2026-111357 Exploitation Priority Update

CVE / Vulnerability Threat Intelligence Report

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1. Executive Summary

CVE-2026-111357 is an Ivanti Connect Secure (ICS) privilege-escalation component that CISA added to the Known Exploited Vulnerabilities (KEV) Catalog on 2026-10-30. CISA's current alert states that it was exploited as part of a chain with CVE-2026-104935 and CVE-2026-118527. Internet-facing ICS appliances must be treated as a high-priority incident-response and patch-validation population.

This report does not characterize CVE-2026-111357 as a standalone unauthenticated entry point. Ivanti's advisory is the authority for chain preconditions, affected builds, and fixed releases. The operational priority is to inventory exposed appliances, apply vendor guidance, run the Integrity Checker Tool, preserve relevant evidence, and assume that successful chain use could yield appliance-level privilege.

2. Vulnerability Metadata

Field	Value
CVE	CVE-2026-111357
Product	Ivanti Connect Secure
CISA weakness label	Missing Authentication for Critical Function
Role in reported activity	Privilege-escalation component in a reported chained exploitation sequence
KEV status	Added 2026-10-30
Exploitation status	CISA and Ivanti describe limited customer impact when the related CVEs are chained
Primary remediation	Apply Ivanti's current fixed release guidance and validate appliance integrity
Important limitation	Do not infer independent unauthenticated exploitation from the KEV entry alone

3. Source Review & Confidence

Source	Contribution	Confidence
CISA KEV catalog	Current priority and date of KEV addition	High
CISA Ivanti alert	Exploited-chain and immediate-action context	High
Ivanti security advisory	Product scope, fixed-release and integrity-check guidance	High
Ivanti October security update	Vendor operational remediation reinforcement	High

The KEV listing establishes prioritization, not every technical prerequisite. No exploit reproduction was performed. Claims about full compromise, actor attribution, or a specific appliance's exposure require environment-specific evidence.

4. Vulnerability Details

CISA labels the issue as a missing-authentication critical-function weakness. Current public guidance places it in an exploitation chain involving related ICS flaws. Connect Secure commonly serves as a remote-access trust boundary; appliance compromise can therefore endanger VPN authentication, administrative access, session material, and downstream internal reachability. Treat the appliance and its management plane as security-critical infrastructure.

5. Attack Scenarios

Scenario	Preconditions	Potential outcome	Priority action
Reported chained exploitation	Vulnerable ICS and documented related chain conditions	Elevated appliance privileges	Patch, run integrity checks, and investigate
Appliance compromise followed by VPN abuse	Access to appliance controls, sessions, or credentials	Lateral movement through remote-access trust	Review authentication, admin, and VPN telemetry
Delayed remediation	Externally reachable vulnerable appliance remains online	Continued scanning/exploitation opportunity	Restrict exposure during maintenance

6. Affected Products and Exposure

Exposure pattern	Why it matters	Required action
Public ICS gateway	Direct exposure to a KEV-listed chain component	Identify version; follow Ivanti remediation immediately
ICS behind approved remote access	Reduced exposure but still a trust boundary	Patch on emergency schedule and review authentication events
Retired but reachable appliance	Often lacks patch ownership and monitoring	Isolate or decommission; validate DNS and firewall records
HA cluster / DR appliance	Drift can leave a vulnerable failover path	Validate every node and test failover after patching

7. Mitigation and Workarounds

Priority	Owner	Action	Verification
Critical	Network / IAM	Inventory every ICS appliance, role, version, and internet exposure	Signed inventory and firewall review
Critical	Platform	Apply the Ivanti advisory's fixed-release guidance	Version and maintenance record
Critical	Incident response	Run the Ivanti Integrity Checker Tool and preserve alerts/logs	Tool output retained with case evidence
High	SOC	Review administrative, authentication, configuration, and outbound-network activity	Hunt results documented

High	IAM	Rotate privileged appliance/admin credentials when integrity concerns exist	Rotation audit complete

8. Detection Engineering

Signal	Telemetry source	Analytic goal
New or unusual ICS administrator activity	ICS audit / identity logs	Identify management-plane manipulation
Authentication successes from novel networks	VPN / IdP logs	Identify abnormal remote-access use
Appliance-originated outbound traffic	Firewall, proxy, NetFlow	Identify unexpected callbacks or staging

```
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where DeviceName has_any ("ivanti", "ics", "pulse")
| where InitiatingProcessFileName !in~ ("", "expected-management-agent")
| project Timestamp, DeviceName, RemoteUrl, RemoteIP, RemotePort, InitiatingProcessFileName
| order by Timestamp desc
```

```
index=vpn (sourcetype=ivanti* OR host=*ics*)
| stats count values(user) as users values(src_ip) as source_ips by action, dest, _time
| sort - _time
```

9. Threat Hunting

Hypothesis: a vulnerable or previously exposed ICS appliance was used to alter access controls or enable suspicious remote-access activity after chain exploitation.

```
SigninLogs
| where TimeGenerated > ago(30d)
| where AppDisplayName has_any ("Ivanti", "Connect Secure", "VPN")
| summarize users=dcount(UserPrincipalName), ips=make_set(IPAddress, 20),
resultCodes=make_set(ResultType, 10) by bin(TimeGenerated, 1h), AppDisplayName
| where users > 5 or array_length(ips) > 10
| order by TimeGenerated desc
```

Triage results by correlating source IPs with ICS audit changes, appliance integrity-check results, and privileged-group modifications. A detection hit is investigation evidence, not proof of exploitation.

10. MITRE ATT&CK Mapping

Technique	Name	Rationale
T1190	Exploit Public-Facing Application	Relevant to exposed remote-access appliances and the reported chain
T1068	Exploitation for Privilege Escalation	Relevant to the reported privilege-escalation component
T1098	Account Manipulation	Hunt for access or admin changes after suspected compromise

11. Validation / Lab Testing

No exploit reproduction or intrusive appliance testing was performed. QA verified the CISA KEV entry, CISA alert, vendor remediation framing, structured tables, embedded KQL/SPL queries, PDF rendering, and extracted text. Operators must validate actual installed versions and maintenance outcomes under approved change control.

12. Recommended Actions Summary

Priority	Action
1	Treat internet-facing ICS as an emergency inventory and remediation target.
2	Apply Ivanti's fixed-release guidance to every production, HA, DR, and management-plane node.
3	Run the Integrity Checker Tool and preserve its output before and after remediation.
4	Hunt for anomalous remote access, appliance administration, and outbound communication; rotate credentials if compromise indicators exist.

13. References

- [1] CISA, Known Exploited Vulnerabilities Catalog. https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json
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- [3] Ivanti, Security Advisory Ivanti Connect Secure (CVE-2026-104935). https://hub.ivanti.com/s/article/Security-Advisory-Ivanti-Connect-Secure-CVE-2026-104935
- [4] Ivanti, October 2026 Security Update. https://www.ivanti.com/blog/october-2026-security-update